Ahmedabad University

Group Members

Devshree

Hiranshee

Prakshal

Yagnik

Dhruv

s2_g5_net

Stochastic Optimization of Spatial Throughput in Wireless Networks under Determinantal Scheduling

Project Context and Motivation

- **Domain**: Wireless Telecommunications and Network Planning.

- **Why Uncertainty Exists:** Radio waves interact unpredictably with the physical environment, creating unavoidable signal fading and interference.

- **Why is uncertainty intrinsic?**
  - Signals fade unpredictably due to physical environments .
  - Device positions are unpredictable, creating chaotic, constantly changing interference patterns.

Project Objective and Scope Main Goal:

- **Exact Objective:** Find the optimal node density λ to maximise network throughp
- **Deterministic Strategy:**
  - Simulating a fixed-grid spatial deployment (Square and Hexagonal) with zero randomized node placement.
- **Why it's Reasonable:**
  - Traditional cellular networks are deployed in planned, rigid grids.

Project System Overview Step-by-step system:

- **Architecture:**

  A Monte Carlo simulation engine evaluating fixed geometries.

- **Data Flow**: Grid Generation → Distance Calculation → Fading Application → SINR Evaluation → Throughput Output.

- **Where Uncertainty Enters**: Specifically at the "Fading Application" stage (the physics), while geometry remains fixed

Sources of Uncertainty in the Project

- **Variability:** Unpredictable signal bounces (Rayleigh fading).
- **Noise:** Aggregate interference from all active transmitters in the city.
- **How it is Simulated:** By applying random exponential multipliers to deterministic path loss formulas.

Key Random Variables and Their Roles

- **Fading (H): Definition:** Environmental signal degradation.
  - **Distribution:** Exponential.
  - **Role:** Modifies raw received power.
- **Interference (I): Definition:** Sum of background noise.
  - **Distribution:** Derived sum.
  - **Role:** Creates the threshold barrier for success.

Probabilistic Modeling with Deterministic Decisions

- **Mathematical Assumptions**: Minimum 2D spatial guardrail (0.02) to prevent infinite signal scaling.

- **Interaction**: Unpredictable fading determines if the fixed decision rule passes or fails.

- **Relevant Equation**: SINR = Signal / (Interference + Noise).

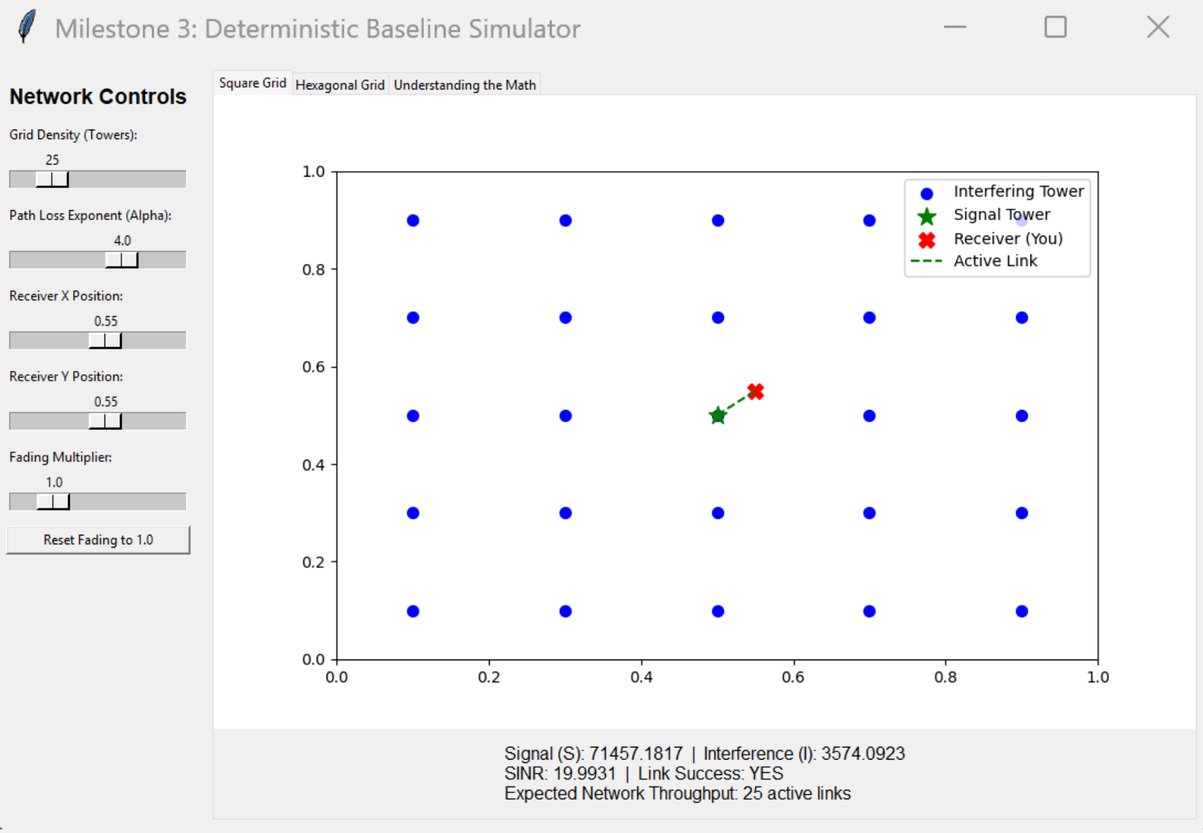
Deterministic Baseline Logic

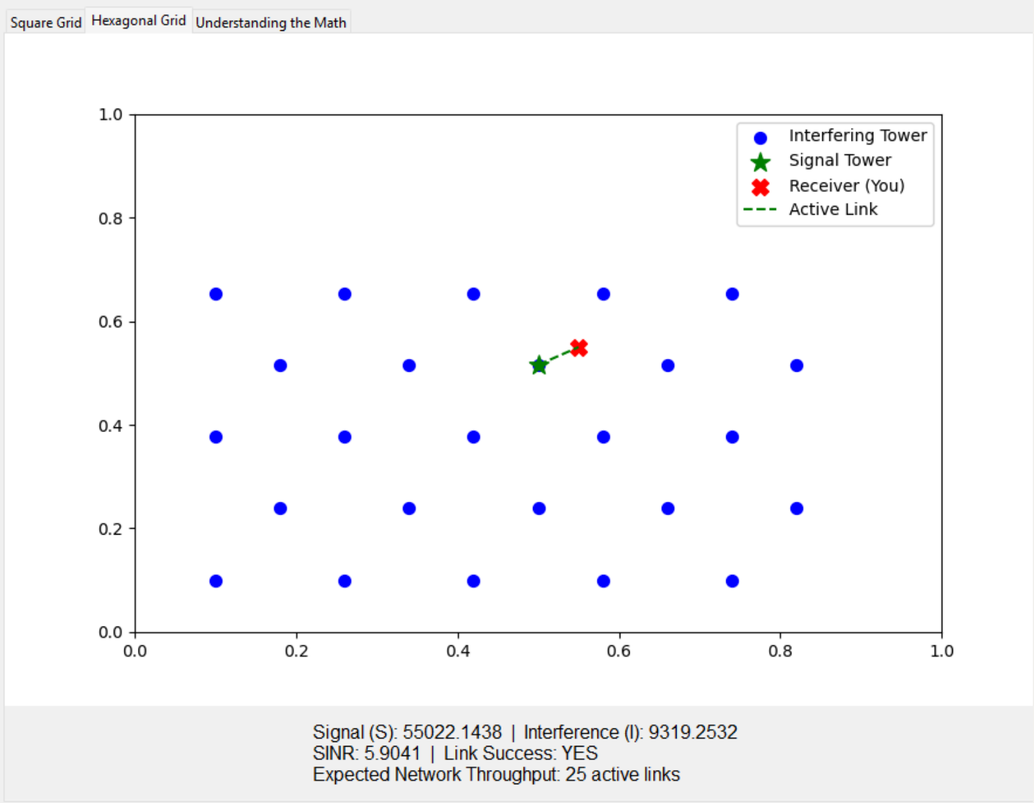
- **Fixed Strategy:** 400 static receiver points testing a rigid grid of towers.

- **Step-by-Step:** Calculate distances ---> Apply fixed threshold ---> Average the successes.

- **Strict Rule:** No randomized scheduling algorithms (No Aloha, No DPP).

Visual Representation of Tower and Receiver





Simulation Implementation

- **Algorithms Used**: Deterministic mesh grids (numpy.meshgrid) for both towers and receivers.
- **Code Structure:** Nested Monte Carlo loops sweeping through increasing densities.
- **Execution:** 50 fading snapshots executed at every single test point to find the true spatial average.

Results and Performance Analysis

- **Metrics:** Spatial Throughput (T) vs. Network Density (lambda).

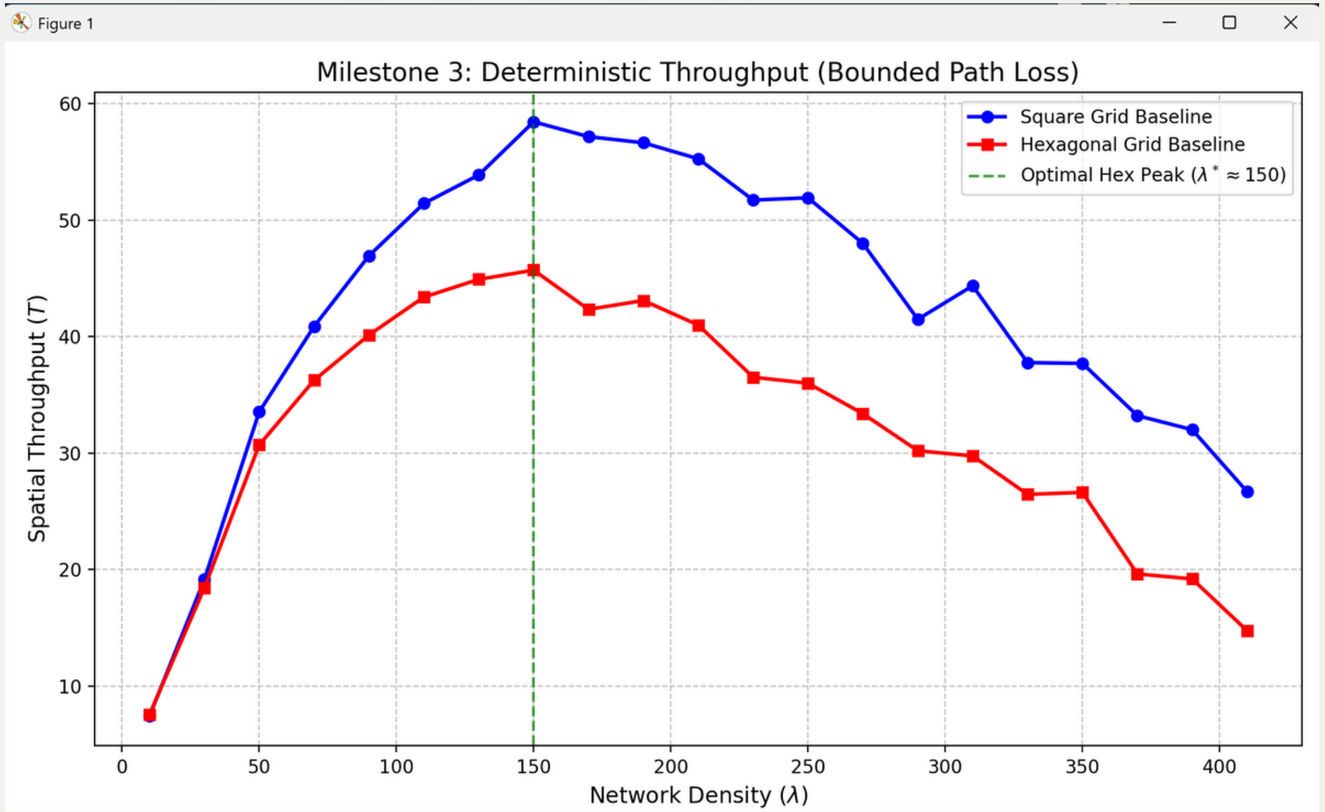
- **Plots:** The Baseline Throughput "Bell Curve".

- **Observed Behavior:** Throughput scales linearly, hits an optimal peak (lambda^*), and crashes under interference.

Square Vs Hexagonal Bell Curve

```
# --- SIMULATION PARAMETERS ---
N_FADING_SIMS = 50            # Fading
ALPHA = 4.0                   # Path l
THRESHOLD = 1.0               # SINR t
ANTENNA_HEIGHT = 0.05         # THE FI
DENSITIES = np.arange(10, 420, 20) #
```

```
Running Bounded Path Loss Monte Carlo Simulation...
Target Density 10 processed. (Actual Sq: 9, Hex: 9)
Target Density 30 processed. (Actual Sq: 25, Hex: 25)
Target Density 50 processed. (Actual Sq: 49, Hex: 49)
Target Density 70 processed. (Actual Sq: 64, Hex: 64)
Target Density 90 processed. (Actual Sq: 81, Hex: 81)
Target Density 110 processed. (Actual Sq: 100, Hex: 100)
Target Density 130 processed. (Actual Sq: 121, Hex: 121)
Target Density 150 processed. (Actual Sq: 144, Hex: 144)
Target Density 170 processed. (Actual Sq: 169, Hex: 169)
Target Density 190 processed. (Actual Sq: 169, Hex: 169)
Target Density 210 processed. (Actual Sq: 196, Hex: 196)
Target Density 230 processed. (Actual Sq: 225, Hex: 225)
Target Density 250 processed. (Actual Sq: 225, Hex: 225)
Target Density 270 processed. (Actual Sq: 256, Hex: 256)
Target Density 290 processed. (Actual Sq: 289, Hex: 289)
Target Density 310 processed. (Actual Sq: 289, Hex: 289)
Target Density 330 processed. (Actual Sq: 324, Hex: 324)
Target Density 350 processed. (Actual Sq: 324, Hex: 324)
Target Density 370 processed. (Actual Sq: 361, Hex: 361)
Target Density 390 processed. (Actual Sq: 361, Hex: 361)
Target Density 410 processed. (Actual Sq: 400, Hex: 400)
```



Ahmedabad University

Limitations of Deterministic Strategy

- **Weaknesses:** Cannot model unpredictable human clustering or movement.
- **Boundary Anomaly:** Square grids artificially beat Hexagonal grids due to edge-effects in our bounded 1×1 simulation window.
- **Poor Performance:** Fails to represent highly dynamic, real-world urban environments.

Summary and Transition to Randomized Strategies

- **Key Insights:** Even perfectly spaced networks have a mathematical breaking point due to interference.

- **Why Randomization is Next:** We must model the true chaos of real users (Aloha) and apply smart, repulsive scheduling algorithms (DPP).

- **The Goal:** Beat this deterministic baseline's peak throughput in Milestone 4 and 5.